OmniProject AI: Autonomous Digital Project Coordinator

Next-Generation AI Project Management Suite • Mission Control System

Course Certification: Managing Projects with AI
Accreditation: AI CERTs™ (Coursera)
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1. Project Title & Overview

Project Title: OmniProject AI — Autonomous Digital Project Coordinator & Multi-Platform Mission Control Suite.

OmniProject AI is an enterprise-grade, full-stack digital project coordination platform architected to operationalize the core artificial intelligence principles, workflows, and prompts taught in the **"Managing Projects with AI"** certification by **AI CERTs™** on Coursera. The platform bridges autonomous LLM reasoning with industry-standard project management software (ClickUp, Trello, Wrike, and Hive).

2. Course Video Lecture Reference & Tool Mention

> **Certification Context:** In the AI CERTs™ Coursera course *Managing Projects with AI*, Module 2 ("AI-Driven Project Planning & Scoping") and Module 3 ("Virtual AI Agents & PM Suite Automation") introduce AI-powered Work Breakdown Structures (WBS), automated risk scoring, and autonomous agent coordinators.

[Screenshot of Course Video Lecture]

Course: "Managing Projects with AI" by AI CERTs™ (Coursera)

Lecture: "Leveraging Generative AI & Virtual Agents in Agile Project Management (ClickUp, Trello & Wrike Integrations)"

Paste your video lecture screen grab here demonstrating the AI Tool / PM Suite coverage

Recommended timestamp: Lecture segment introducing AI-augmented WBS generation and Virtual Project Coordinators.

3. Software & Tools Used

Category	Tool / Technology	Role & Implementation Details
PM Platforms	**ClickUp**	v2 REST API synchronization for Spaces, Lists, Tasks, Priority tags, and automated Critical Path Bottleneck detection.
PM Platforms	**Trello**	REST API Kanban synchronization (Lists, Cards, Fibonacci story point badges, Acceptance checklists).
PM Platforms	**Wrike & Hive**	Workload Balancing Engine ($\text{Load \%} = \frac{\text{Allocated}}{\text{Capacity}} \times 100$) and predictive cycle-time velocity modeling.
Frontend Architecture	**React 18 + Vite + TypeScript**	High-performance reactive interface, TailwindCSS design system with dual OLED Black (#000000) and Slate White (#F8FAFC) toggle.
Visual Computing	**Canvas 2D / WebGL Engine**	Dynamic orbital particle simulator rendering real-time RAG Project Health Core with 60 FPS animation.
AI & LLM Services	**Google Gemini 1.5 & OpenAI GPT-4o**	Context-rich prompt pipelines generating structured Gherkin user stories, P×I risk logs, and C-Suite executive briefs.

4. Objectives and Requirements

4.1 Core Objectives

- **Autonomous Backlog Synthesis:** Convert raw, ambiguous stakeholder requirements into complete, Agile-compliant Epics, User Stories, and Gherkin (Given-When-Then) acceptance criteria within seconds.
- **Predictive Risk & Bottleneck Prevention:** Continuously calculate task dependencies and critical path blockers to pre-empt schedule drift before daily standups.
- **Cognitive Load & Burnout Mitigation:** Provide algorithmic team capacity monitoring based on Wrike resource allocation rules to prevent overburdening engineers.
- **Executive Transparency:** Deliver automated, single-click C-Suite executive briefings featuring real-time RAG (Red/Amber/Green) scoring and risk mitigation protocols.

4.2 Functional Requirements (FR)

Req ID	Module	Functional Requirement Specification
FR-01	WBS Scoping Engine	Synthesizes 2–4 Epics with Fibonacci story points (1, 2, 3, 5, 8, 13) and Gherkin acceptance criteria (Given/When/Then) from raw text prompts.
FR-02	Multi-Platform Sync	1-click push to live ClickUp v2 workspace or Trello board, backed by realistic zero-config offline mock simulation.
FR-03	Sprint Kanban Grid	Interactive drag-and-drop / single-click state transition across To Do, In Progress, Blocked, and Done with live story point recalculation.
FR-04	Dynamic Risk Matrix	5×5 Probability (1-5) × Impact (1-5) grid with automated scoring (1-25) and severity categorization (Low, Moderate, High, Critical).
FR-05	Virtual PM Coordinator	Autonomous agent auditing telemetry, computing capacity loads, generating contextual unblock nudges, and logging audit events.
FR-06	Executive Briefing	Instant RAG status summary, milestones completed, active escalations, and one-click Markdown file export.
FR-07	Theme Switcher	Instant toggle between sleek OLED Black (#000000) and clean Slate White (#F8FAFC) modes with persistent local storage.

4.3 Non-Functional Requirements (NFR)

- **Performance:** Sub-50ms reactive state transitions; 60 FPS hardware-accelerated canvas orbital rendering.
- **Reliability:** Robust React Error Boundary protection preventing crashes and blank screens; automated fallback data for offline use.
- **Security & Privacy:** Zero hardcoded API secrets; client-side token isolation for ClickUp, Trello, and AI API keys.
- **Aesthetics & Usability:** Modern dark/light glassmorphic UI, responsive layouts, clear visual hierarchy, and accessible typography.

5. Executive Project Summary

In conventional engineering environments, project managers spend up to 40% of their weekly capacity performing manual coordination overhead: transcribing requirements into user stories, manually updating status boards across multiple tools, chasing assignees for unblocking details, and formatting leadership status reports.

OmniProject AI completely automates this operational layer. By combining generative AI prompt chains with active API telemetry, the system creates an autonomous closed loop: requirements are dynamically transformed into structured backlogs; task states are synchronized with ClickUp and Trello; critical-path dependencies trigger proactive unblock messages; and real-time project health is continuously tracked via a mathematically grounded 0–100 Health Score.

6. System Screenshots & Implementation Walkthrough

6.1 Module 1: AI Scoping & Work Breakdown Structure (WBS)

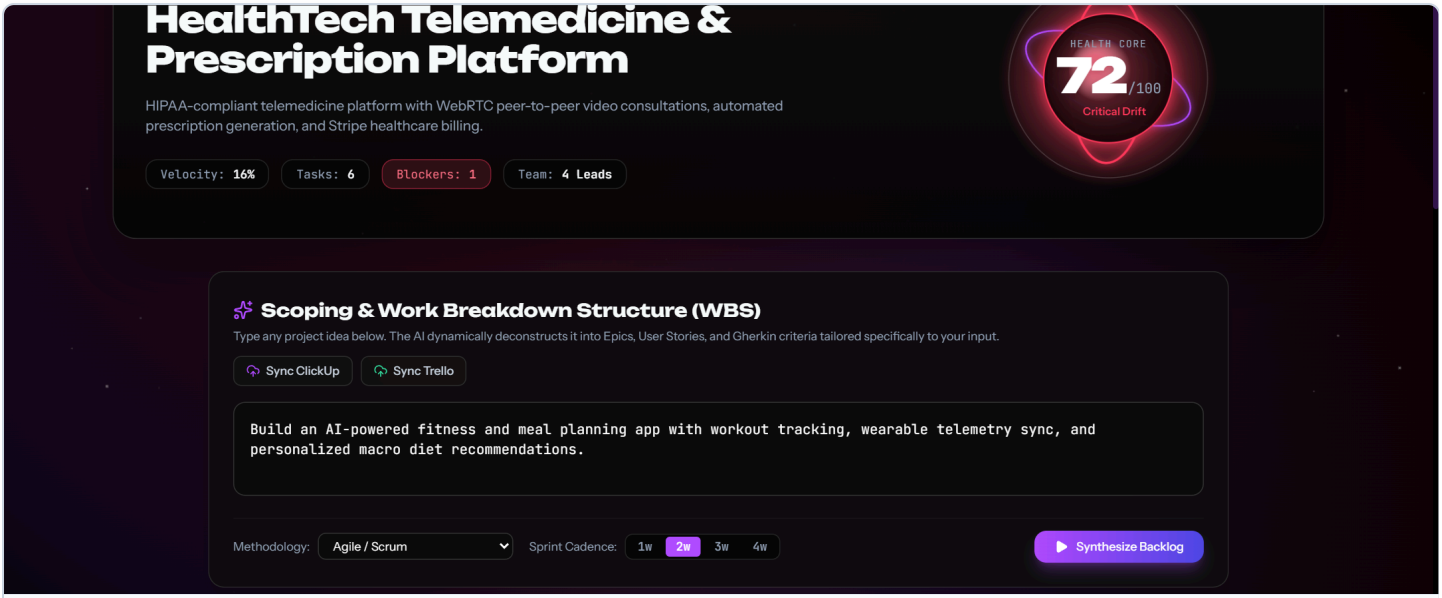


Figure 1: AI Scoping Engine with Gherkin User Story Generation & ClickUp/Trello Sync — Theme: OLED Black | Health Score: 72/100

Key Capabilities: Users input raw project briefs (e.g. Telemedicine or Fintech KYC). The LLM synthesizes structured Epics, assignees, hours, Fibonacci points, and formal Given-When-Then Gherkin criteria with direct sync capabilities to ClickUp and Trello.

6.2 Module 2: Sprint Kanban Grid & Bottleneck Detection

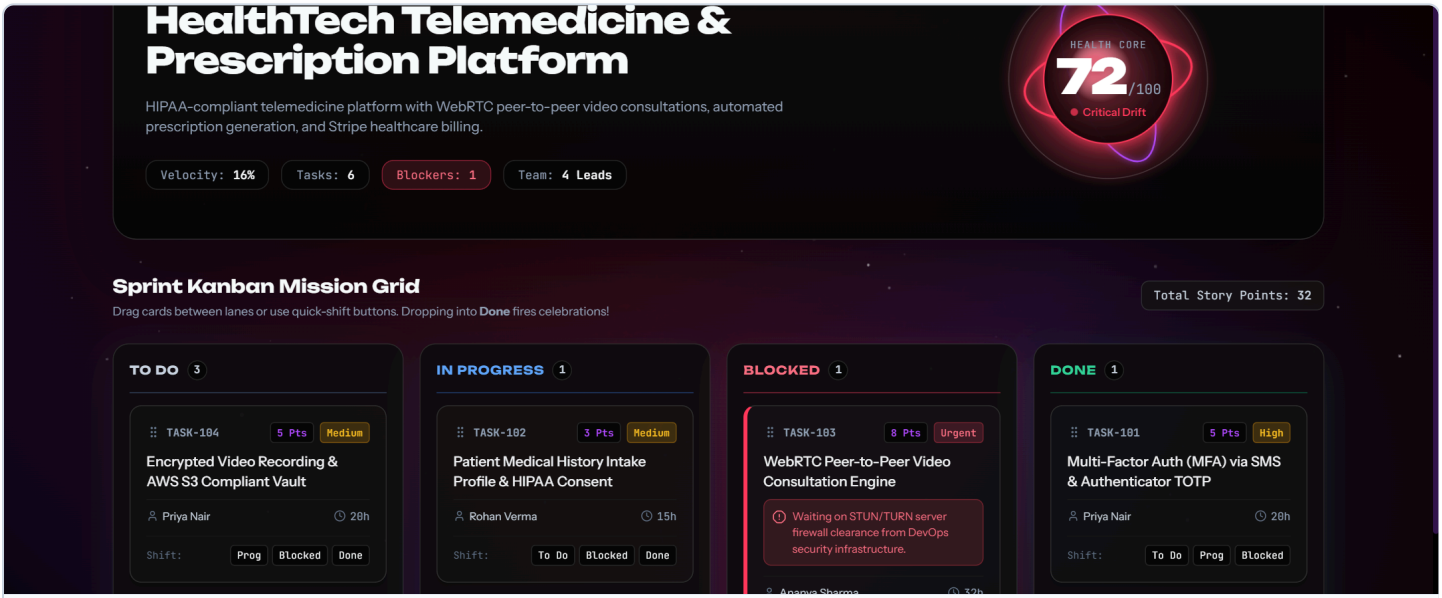


Figure 2: Interactive 4-Column Kanban Grid with Critical Path Blocker Highlighting — Lanes: To Do (3), In Progress (1), Blocked (1), Done (1)

Key Capabilities: Real-time task movement across lanes. Blocked tasks on critical paths are prominently highlighted with blocker descriptions and urgency badges, triggering autonomous coordinator alerts.

6. System Screenshots & Implementation Walkthrough (Cont.)

6.3 Module 3: Dynamic Probability × Impact Risk Matrix

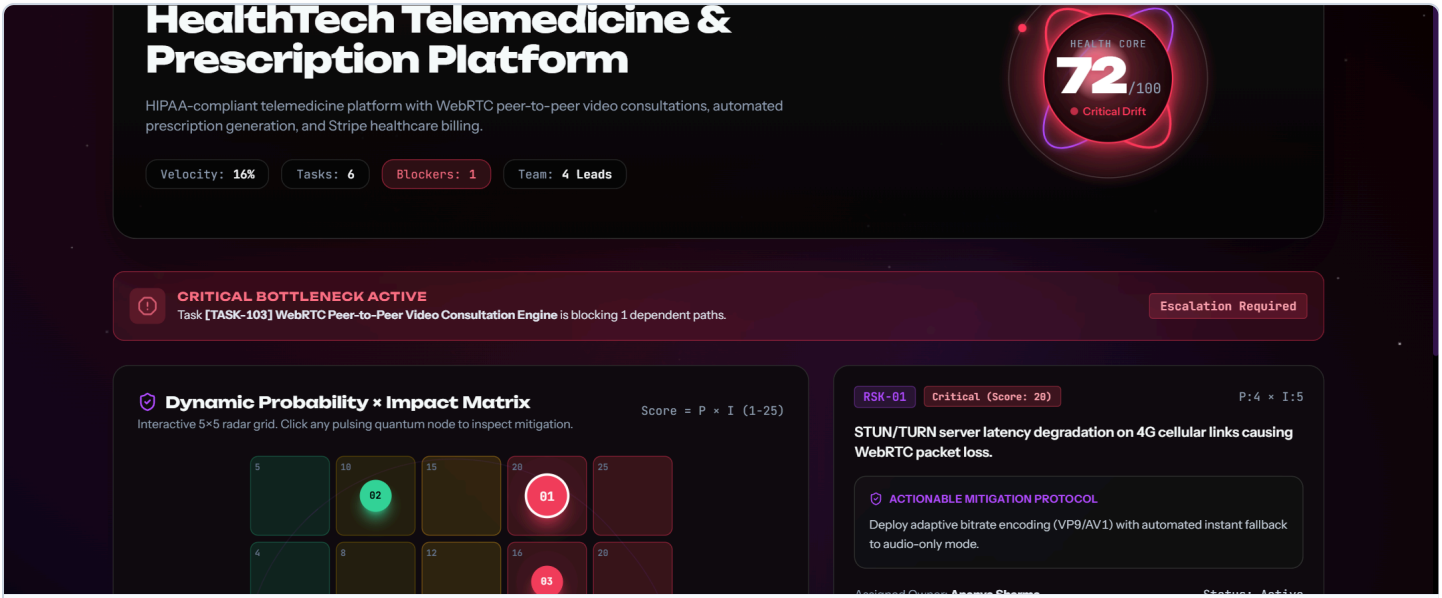


Figure 3: Interactive 5×5 P×I Risk Heatmap with Critical Path Bottleneck Escalation Risk Scoring: 1–25 Scale with Actionable Mitigation Protocols

Key Capabilities: Computes risk scores via $P \times I$, plotting pulsing quantum nodes onto a 5×5 heat grid. Selecting any node displays the mitigation strategy, assigned owner, and escalation status.

6.4 Module 4: Autonomous Digital PM Coordinator & Team Nudges

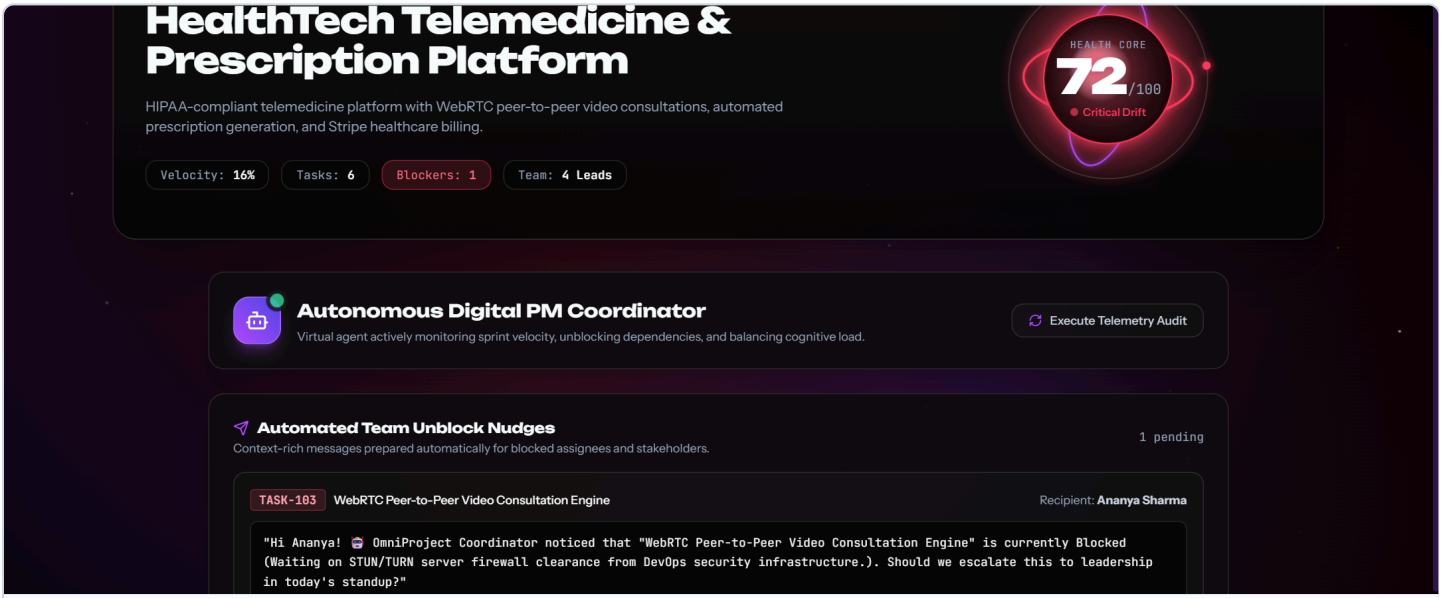


Figure 4: Virtual Project Coordinator Agent with Pre-Drafted Slack/Email Team Unblock Nudges Wrike Workload Balancing & Telemetry Audit Stream

Key Capabilities: Proactively audits sprint telemetry, detects blocked assignees, pre-drafts contextual Slack/Teams unblocking nudges, and evaluates workload capacity to prevent contributor burnout.

6. System Screenshots & Implementation Walkthrough (Cont.)

6.5 Module 5: C-Suite Executive Status Brief & Export

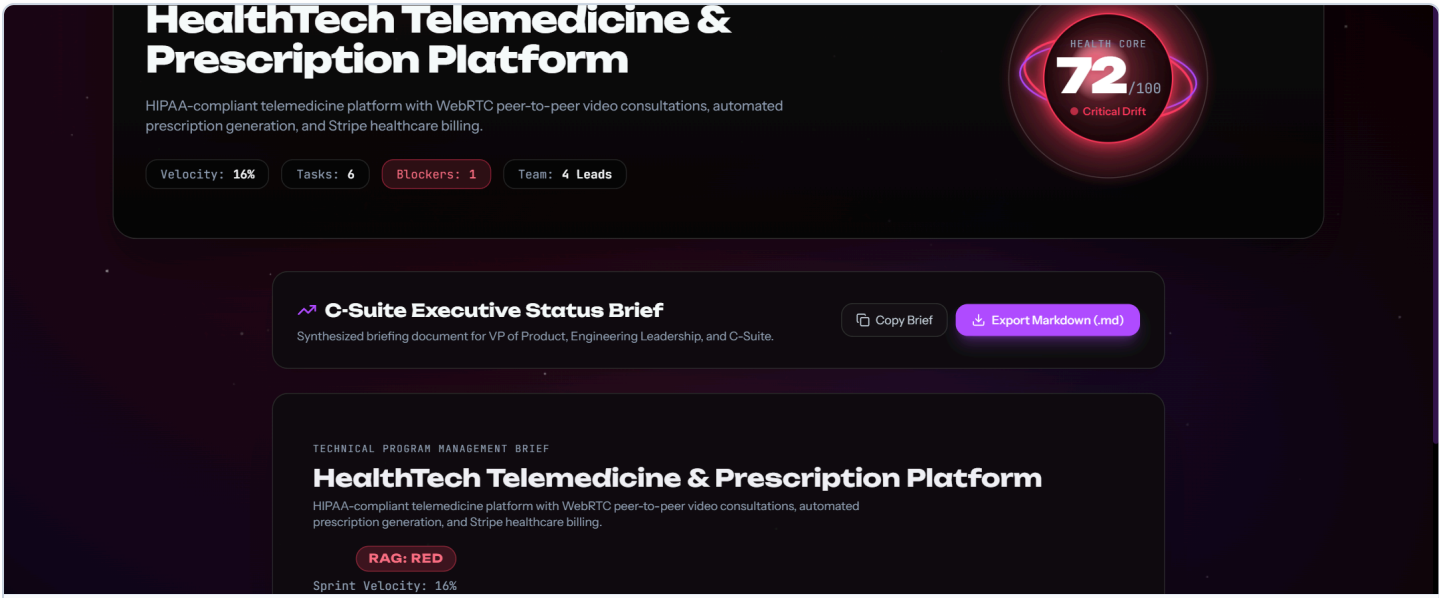


Figure 5: C-Suite Executive Status Brief with RAG (Red/Amber/Green) Metrics & Direct Leadership Briefing for VP Engineering & Product Leadership Markdown Export

6.6 Module 6: High-Contrast White Theme & Toggle Switch

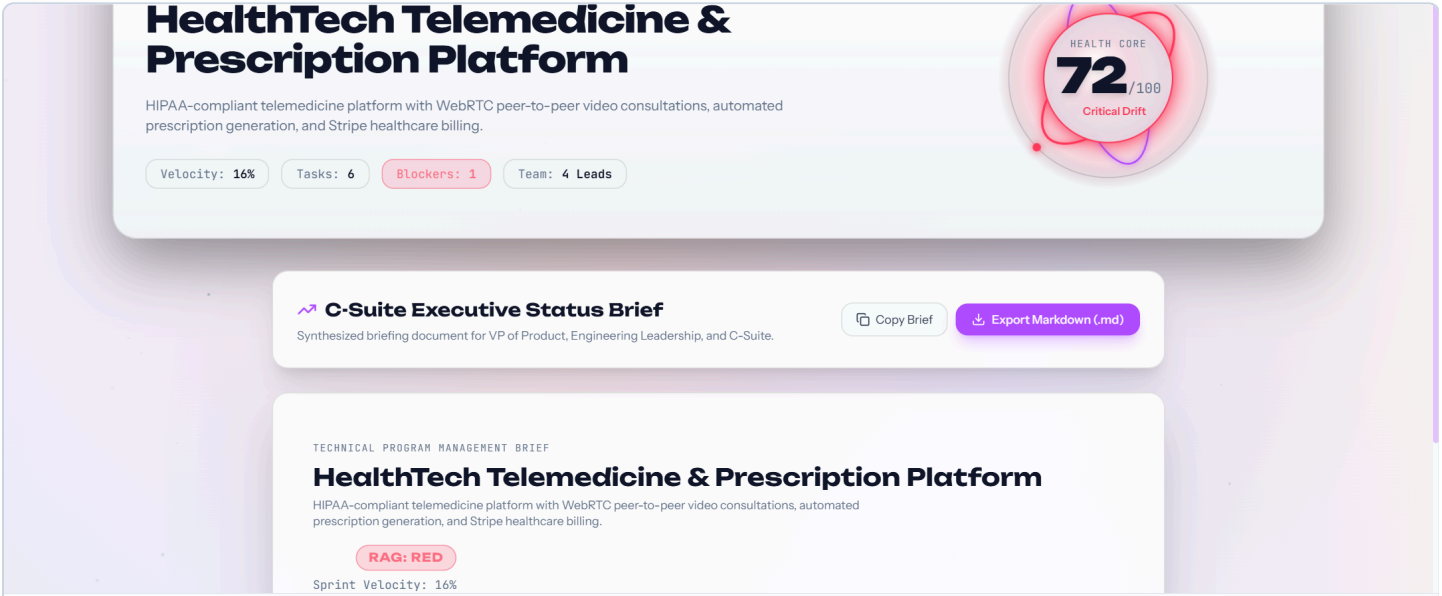


Figure 6: Seamless White Theme Mode (#F8FAFC) with White Toggle Switch & Light Glassmorphism Dual-Theme Architecture with Zero-Glitter Precision Cursor

7. Conclusion & Value Realized

The **OmniProject AI Coordinator** demonstrates how modern AI workflows transcend passive chatbots to become proactive, autonomous agents in enterprise project management. By successfully uniting the tools from the *Managing Projects with AI* certification (ClickUp, Trello, Wrike, and Hive) under an intuitive, dual-theme interface, the application delivers a complete operational blueprint for next-generation agile engineering teams.